GREENING INTERIOR SURFACES


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Jim LaRue founded The HouseMender, Incorporated, and is now in active retirement as a green building consultant in the Cleveland, Ohio, area and nationally.

When we home energy folks decided to identify "building performance" as a description for our work, little did we know what a Pandora's box we opened. This identification now requires us to look at everything that affects a structure's total performance. And yes, that includes the interior surface materials we use in those structures. Performance related to energy efficiency and comfort has now expanded into the kinds of stuff we use to finish the structure. Even if your particular work is focused on energy matters, it requires you to have a clear sense of what is going on in the rest of the structure, because inevitably there will be a connection between them.

Early on, our energy work generated questions about moisture control, and that led to mold, which was an indoor air quality issue. Then we discovered that our heat ducts could be the transport mechanism for much of the mold found in our basements and crawlspaces. That leads to the focus of this article. (The presence of basement mold was the most critical finding in the Healthy House program we developed here in Cleveland after the death of infants suffering from bleeding lung disease; mold spores in terribly damp and moldy basements were coming directly into the children's living space via the ductwork.)

Everything we do in the home has a performance dimension, and I have always assumed that the very use of the word "performance" guarantees our commitment to optimum performance. What is optimum performance? It means creating structures that provide comfortable living space with minimal negative impacts on the resident's environment and the natural environment.

This is not an easy task as it relates to surface materials we may use in our work, because the industries that produce the product are only really just beginning to get the message that we cannot continue to produce product that has negative impacts on personal or natural environments. But we can demand better products by choosing only those products that come closest to meeting our objectives. It is also not easy because the chemistry of products that appear in themselves to be okay may interact negatively with other products. I remember investigating several homes one summer where several residents who had just painted a room were experiencing terrible odors, and surprisingly, it was worse when the windows were open. With the help of some building scientist friends, I learned that the high ozone levels in the outside air from auto emissions during the very hottest days of summer were interacting with a particular mildewcide that several paint companies had put in their interior house paint to avoid potential mold issues.

In another investigation I discovered that the salt a family was putting on sidewalks to break up ice and snow was being tracked into the house. Once inside, the salt interacted with the pesticide that had been sprayed around the baseboards of the house, including the entry area. The odor was unbearable. The safety data sheets developed by manufacturers don't begin to tell the whole story of what can happen when particular products


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The ceramic tiles are made with recycled glass components.


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This countertop is made of wheatboard and no-VOC bonding agents.

are used, because manufacturers are not likely to know all the environments into which their product will enter.

How can we summarize the sensitivity we need to bring to the selection of materials we will be using in our work? Let me suggest the following:

- Carefully review all safety data sheets of products we use (caulks, paints, composite wood products, flooring surfaces, and so on) for any negative effects they could have on the health of the residents and workers, both in the short term and in the long term. Some products will outgas in a very brief time and should not be a hazard to the residents, but worker exposure could be an issue.

- Our work has profound effects on the environment as byproducts from combustion and materials enter the atmosphere. The cumulative effect of all such pollutants (CO_2 in particular) from millions of living spaces in this country alone is contributing to global warming.

- Keep aware of all the nonrenewable resources we are using in our work. Those of us who have entered building performance via the energy field know that the fuels we use (gas and oil) are not renewable. We are constantly trying to find ways to reduce the demand, even as those who are finishing surfaces on the interior are searching for and using renewable products.

- By acting as good stewards of renewables, such as wood, we can promote environmentally responsible practices. Clear-cutting of forests can damage our capacity to grow what we need. Incorporating optimum-value engineering into our work plans can reduce the amount of wood product required. In our first green homes here in Cleveland, we estimated that we used one-third less wood than standard construction, because the houses were designed and the materials were ordered in such a way as to reduce the amount of wood used.

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The cabinets are made with no exposed particleboard edges or surfaces that could outgas and no-VOC finishes.

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This carpet tile is made of recycled materials and is itself nearly 100% recyclable.

- Whenever possible we need to take advantage of recycled or reusable materials. One of my clients used all recycled wood for their floors on the second level and used fly ash in the cement in the concrete floors on the ground level. The client also stained and imprinted a pattern in the concrete surface so the floor would not need additional surface materials.

- Recycling is also essential. A heating contractor friend dismantles each furnace he removes in such a way that he is able to take every bit of the unit to recyclers. He also takes all the cardboard and wood that the new furnace may be packaged in to recyclers.

I have heard some professionals who are adopting building performance describe it as a necessary inconvenience, but there are a growing number who are recognizing this as a growth opportunity and an exciting frontier. It is not easy, but it is necessary and it can be satisfying. It is a Pandora's box worth opening!

For more information:

This article is part of a series on green building that Jim LaRue has prepared for *Home Energy*. Visit our Web site (www.homeenergy.org) to read the complete series.

Your Green Home, by Alex Wilson (Gabriola Island, BC: New Society Publishers, 2006) offers great advice on greening your home. *Environmental Building News*' Green Spec Directory, as well as its Web site, offer the latest information on green products. Both can be found at www.buildinggreen.com.

Check the U.S. Green Building Council Web site to learn what local green building organizations are doing. The site also gives local resources for green building, including surface materials. Go to www.usgbc.org.